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SOFWARE DESIGN & ARCHITECTURE

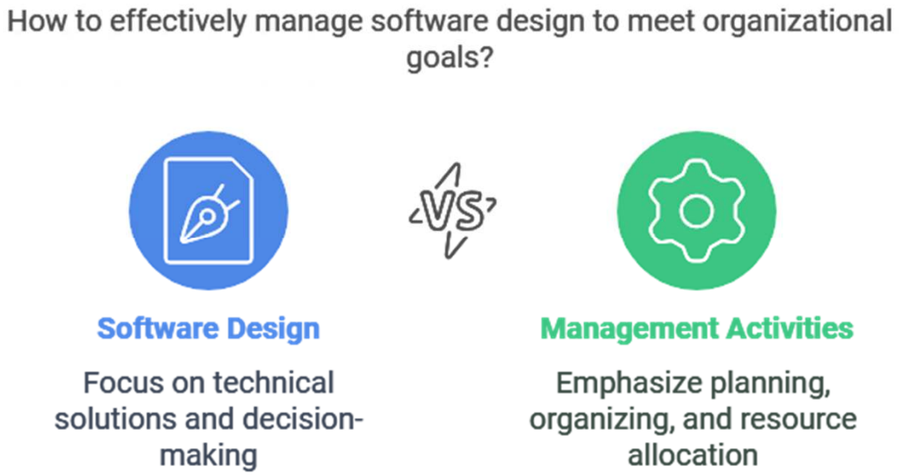
Chapter 9 – Software Design Management

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DESIGN MANAGEMENT FRAMEWORK

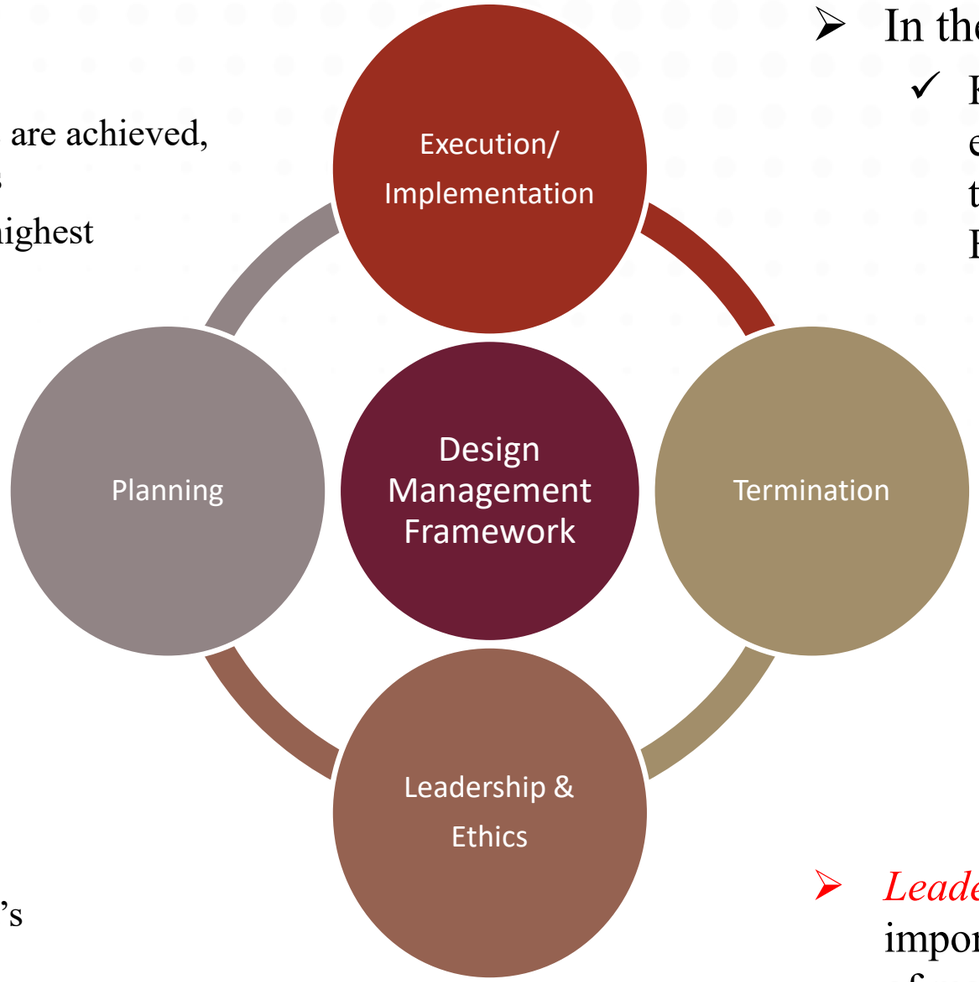
- What is software design management?
  - ✓ Software design is the process of transforming functional and non-functional requirements into models that describe a technical solution.
    - Can be viewed as a complex decision problem since
      - there are many ways to design a software system, and
      - software engineers must make design decisions to achieve the required goals of the software system in the most effective manner
  - ✓ Management is a *set of activities* directed at an organization's *resources* to achieve organizational goals in an efficient and effective manner.
    - Activities: planning and decision making, organizing, leading, and controlling
    - Resources: human, financial, physical, and information



DESIGN MANAGEMENT FRAMEWORK

- The implementation stage is where:
  - ✓ Most of the effort is employed
  - ✓ Most of the design project deliverables are achieved, including completed design documents
  - ✓ The percent project completion is the highest

- In the initial stage, there are two main objectives:
  - ✓ Achieve a clear understanding of the tasks to be accomplished and the resources that are necessary to successfully complete the tasks.
  - ✓ Develop a plan to complete the project's tasks within schedule and budget constraints.

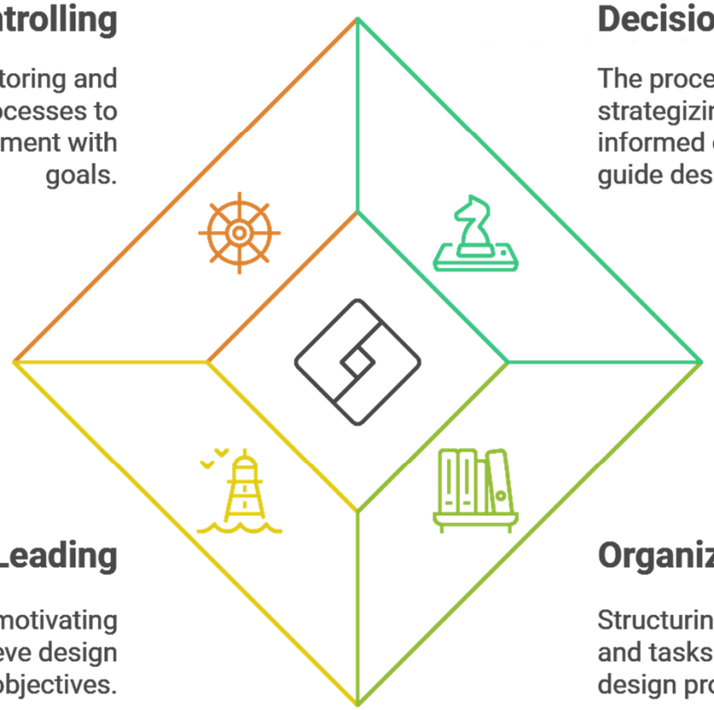


- In the termination stage,
  - ✓ Key activities involve verification that everything is in place for a smooth transition into the code construction phase. Examples of activities are:
    - Making sure that the latest versions of the design documents are securely stored according to configuration management procedures
    - Updating schedule and cost current values
    - Re-evaluating schedule and budget plans based on the resulting performance measures of the design phase
    - Communicating results to upper management.

- *Leadership* and *good ethical conduct* are important factors that affect the effectiveness of managing each lifecycle stage.

DESIGN MANAGEMENT FRAMEWORK

Software Design Management Component



Controlling

Monitoring and adjusting processes to ensure alignment with goals.

Planning & Decision Making

The process of strategizing and making informed choices to guide design.

Leading

Guiding and motivating teams to achieve design objectives.

Organizing

Structuring resources and tasks to streamline design processes.

Achieving Effective Design Management

Mitigate Risks

Identify and address potential challenges to ensure project success.

Engage Stakeholders

Foster active participation and communication among all project participants.

Allocate Resources

Efficiently distribute resources to optimize project execution and outcomes.

Align Goals

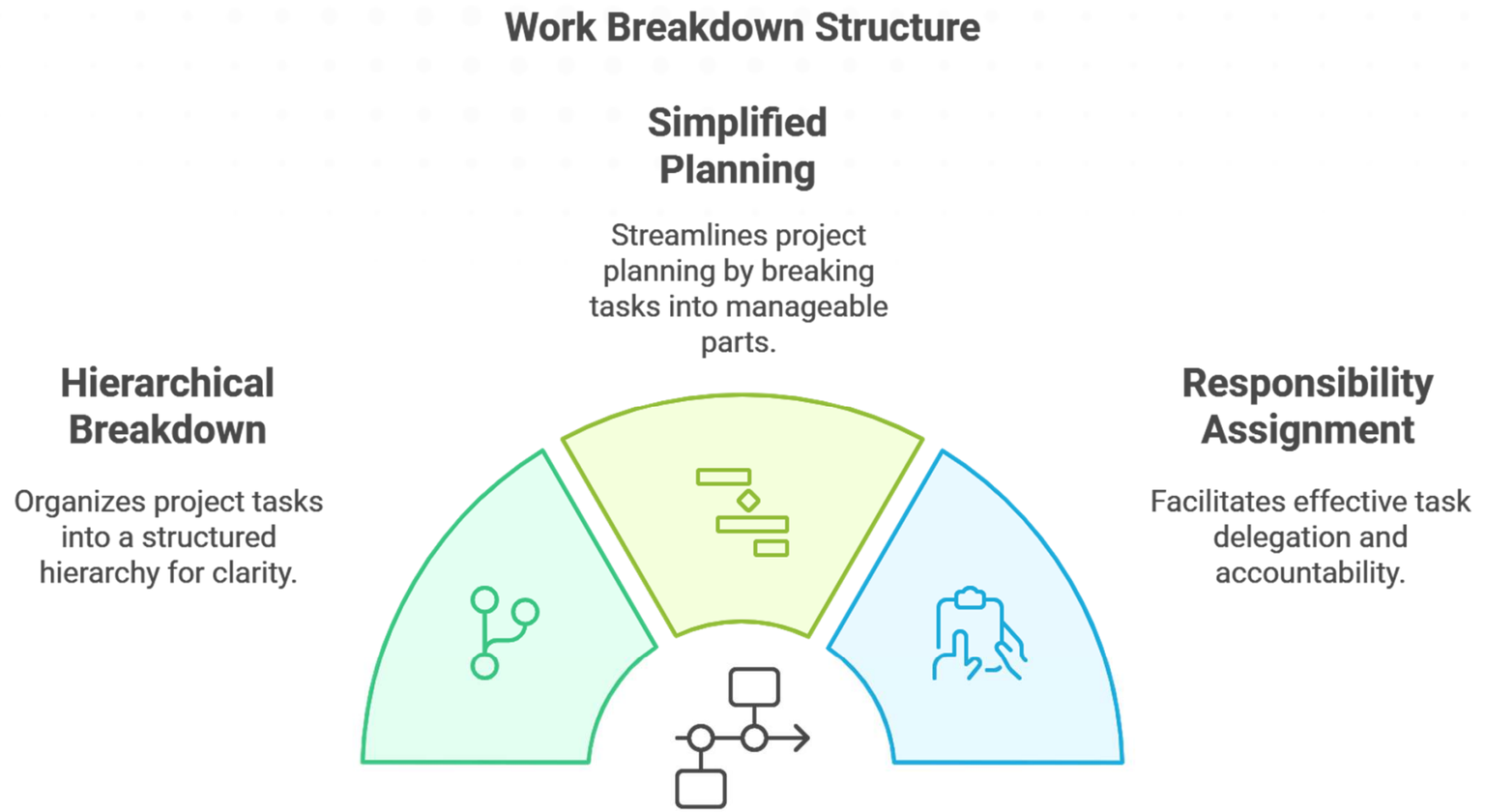
Establish shared objectives among stakeholders to ensure a unified direction.

WORK BREAKDOWN STRUCTURE (WBS)

What is WBS?

•**Definition**:
A hierarchical breakdown of a project into smaller, manageable components.

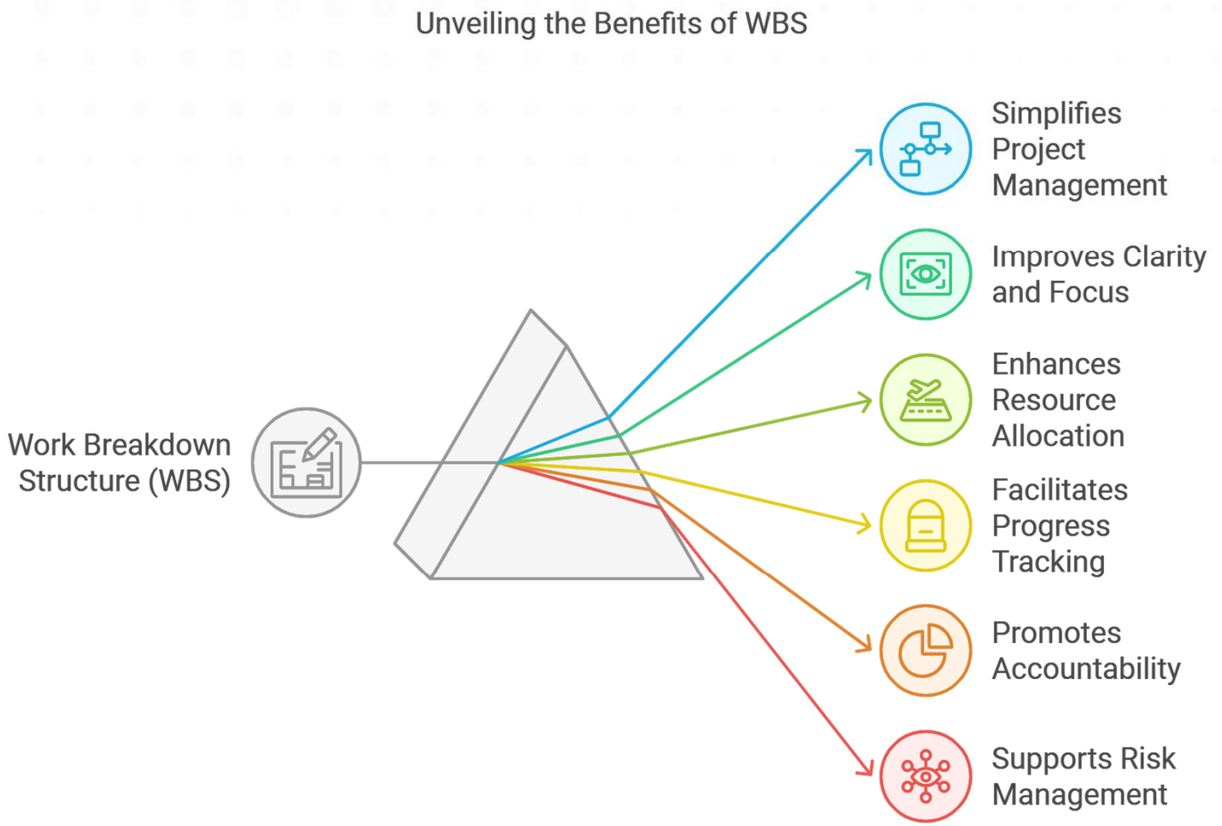
•**Purpose**:
- Simplifies planning and tracking.
- Helps assign responsibilities.



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WORK BREAKDOWN STRUCTURE (WBS)

Outline #	Task Name
1	Design Project
1.1	Architectural Design
1.1.1	Evaluate Alternative Designs
1.1.1.1	Prioritize Objectives
1.1.1.2	Formal Review of Evaluations
1.1.2	Select Architectural Design
1.1.2.1	Make Initial Selection
1.1.2.2	Formal Review to Select Final Design
1.2	Detailed Design
1.2.1	Evaluate Alternatives
1.2.2	Select Among Alternatives
1.3	Documentation
1.3.1	Write Architectural Design Documentation
1.3.2	Formal Review (Architecture Design)
1.3.3	Final Architectural Design Document
1.3.4	Write Detailed Design Documentation
1.3.5	Formal Review (Detailed Design)
1.3.6	Final Architectural Design Document



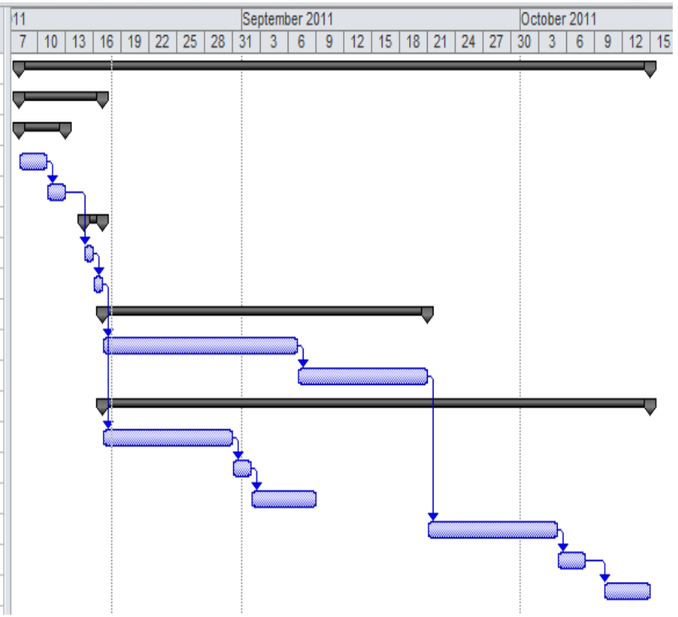
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WORK BREAKDOWN STRUCTURE (WBS)

Outline #	Task Name	Staff_ 1	Staff_ 2	Staff_ 3	Staff_ 4	Staff_ 5
1	Design Project					
1.1	Architectural Design	1	2			3
1.1.1	Evaluate Alternative Designs	1	2			
1.1.1.1	Prioritize Objectives	1	2			
1.1.1.2	Formal Review of Evaluations			1	3	3
1.1.2	Select Architectural Design	1				
1.1.2.1	Make Initial Selection	1				
1.1.2.2	Formal Review to Select Final Design	1		2	3	3
1.2	Detailed Design				1	3
1.2.1	Evaluate Alternatives			2	1	
1.2.2	Select Among Alternatives			2	1	3
1.3	Documentation					
1.3.1	Write Architectural Design Documentation	1	2			
1.3.2	Formal Review (Architecture Design)	1			3	3
1.3.3	Final Architectural Design Document	1		3		
1.3.4	Write Detailed Design Documentation	2			1	3
1.3.5	Formal Review (Detailed Design)	2			1	3
1.3.6	Final Detailed Design Document			3	1	3
Legend:						
1 = Primary responsibility						
2 = Support development role						
3 = Review role						

LINEAR RESPONSIBILITY CHART

	Outline #	Task Name	Duration	Start	Finish	Predecessors
1	**1**	**Design Project**	**50 days?**	**Mon 8/8/11**	**Fri 10/14/11**	
2	**1.1**	**Architectural Design**	**7 days?**	**Mon 8/8/11**	**Tue 8/16/11**	
3	**1.1.1**	**Evaluate Alternative Designs**	**5 days**	**Mon 8/8/11**	**Fri 8/12/11**	
4	1.1.1.1	Prioritize Objectives	3 days	Mon 8/8/11	Wed 8/10/11	
5	1.1.1.2	Formal Review of Evaluations	2 days	Thu 8/11/11	Fri 8/12/11	4
6	**1.1.2**	**Select Architectural Design**	**2 days?**	**Mon 8/15/11**	**Tue 8/16/11**	
7	1.1.2.1	Make Initial Selection	1 day?	Mon 8/15/11	Mon 8/15/11	5
8	1.1.2.2	Formal Review to Select Final Design	1 day?	Tue 8/16/11	Tue 8/16/11	7
9	**1.2**	**Detailed Design**	**25 days**	**Wed 8/17/11**	**Tue 9/20/11**	
10	1.2.1	Evaluate Alternatives	15 days	Wed 8/17/11	Tue 9/6/11	8
11	1.2.2	Select Among Alternatives	10 days	Wed 9/7/11	Tue 9/20/11	10
12	**1.3**	**Documentation**	**43 days**	**Wed 8/17/11**	**Fri 10/14/11**	
13	1.3.1	Write Architectural Design Documentation	10 days	Wed 8/17/11	Tue 8/30/11	8
14	1.3.2	Formal Review (Architecture Design)	2 days	Wed 8/31/11	Thu 9/1/11	13
15	1.3.3	Final Architectural Design Document	5 days	Fri 9/2/11	Thu 9/8/11	14
16	1.3.4	Write Detailed Design Documentation	10 days	Wed 9/21/11	Tue 10/4/11	11
17	1.3.5	Formal Review (Detailed Design)	3 days	Wed 10/5/11	Fri 10/7/11	16
18	1.3.6	Final Architectural Design Document	5 days	Mon 10/10/11	Fri 10/14/11	17

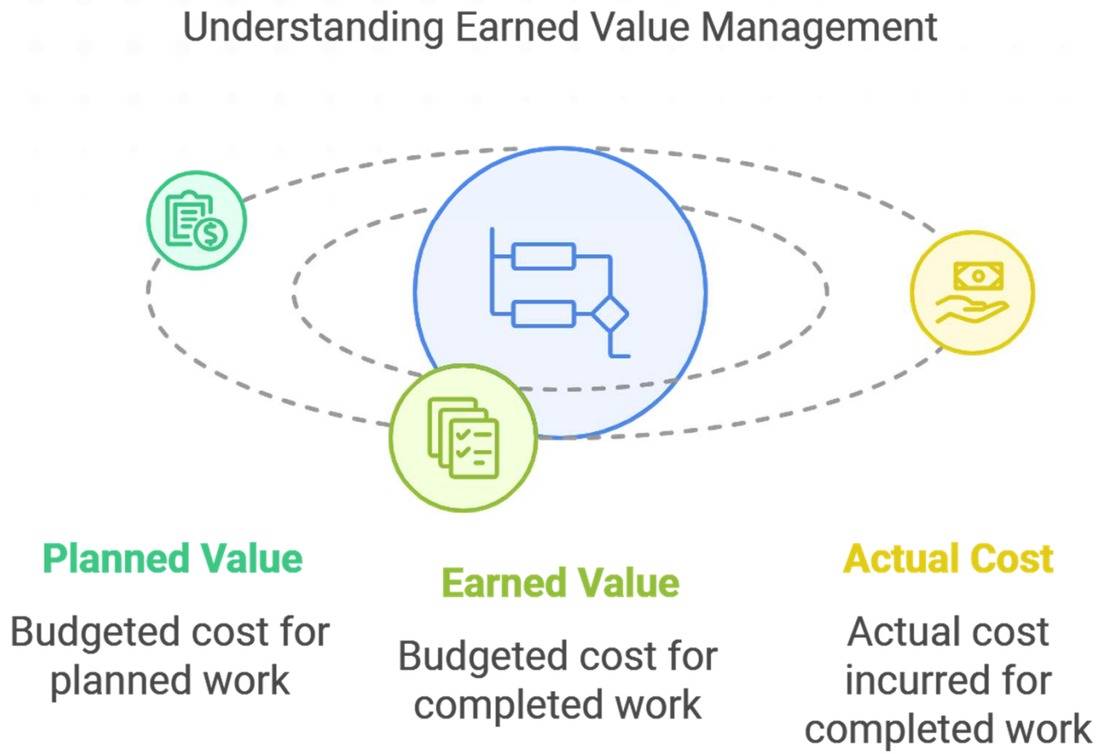


GANTT CHART

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EARN VALUE MANAGEMENT (EVM)

- EVM is a project management technique to determine the progress of tasks based on the value of the work currently completed versus the work that was expected to be completed at that particular time.



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EARN VALUE MANAGEMENT (EVM)

Parameter	Description
EV	Also called *budgeted cost of work performed (BCWP)*, it is the dollar amount of the work currently completed
AC	Also called *actual cost of work performed (ACWP)*, it represents the amount of money that was spent in the work currently completed
PV	Also called *budgeted cost of work scheduled (BCWS)*, it represents the value of the work that was expected to be completed at that point in time of the schedule
CV	Describes the difference between the value of work completed and the actual cost spent to complete it
SV	Describes the difference between the value of work currently completed and the cost of the work that was expected to be completed
CPI	Describes the current cost efficiency of the work (CPI < 1 means over budget; CPI > 1 means under budget; CPI = 1 means right on budget)
SPI	Describes the current schedule efficiency of the work (SPI < 1 means behind schedule; SPI > 1 means ahead of schedule; SPI = 1 means right on schedule)
BAC	Total budget for the project
ETC	Represents the estimated cost remaining to complete the project
EAC	Represents is the total cost of the entire project

EVM DEFINITION PARAMETER

Parameter	Equation
EV	EV = (% work completed) * planned cost for the work
CV	CV = EV - AC
SV	SV = EV - PV
CPI	CPI = EV / AC
SPI	SPI = EV / PV
ETC	ETC = (BAC – EV) / CPI
EAC	EAC = ETC + AC

EVM EQUATION

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EARN VALUE MANAGEMENT (EVM)

Sara is managing a software design project composed of 10 tasks. Each task was estimated to cost $1,000; therefore, BAC = $10,000. Each task was estimated to be completed in a month; therefore, the total planned duration for the project was estimated to be 10 months. Since each task is expected to have the same cost and duration, the completion of an individual task represents 10% completion of the design project. In the current fifth month, only three tasks have been fully completed at a cost of $4,000. Sara needs to provide upper management with current progress status.

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EARN VALUE MANAGEMENT (EVM)

- EV is calculated to be $3,000 given that the three tasks completed represent 30% of the project, and the budget for the entire project is $10,000.

- To calculate SV, we use a PV of $5,000 because each task was budgeted at $1,000 and five tasks should have been completed in the fifth month.

- The shows the calculations for the rest of the EVM parameters.

- These results clearly show that the project is over both budget and schedule estimates.
  - ✓ The CPI index, for example, shows that only $0.75 of earned value was received for every dollar actually spent.
  - ✓ The EAC shows an estimated $6,333 over budget amount.
  - ✓ These alarming values give Sara and upper management the opportunity to implement risk mitigation policies to either get the project to its budgeted path or at least attempt to minimize losses.

Parameter	Calculation
EV	0.3 * $10,000 = $3,000.
CV	$3,000 - $4,000 = -$1,000
SV	$3,000 - $5,000 = -$2,000
CPI	$3,000 / $4,000 = 0.75
SPI	$3,000 / $5,000 = 0.6000
ETC	($10,000 - $3,000) / 0.75 = $9,333
EAC	$9,333 + $7,000 = $16,333

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THANK YOU

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